

The Role of Intestinal Microbiota in Graft versus Host Disease

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Abstract: Graft versus host disease (GVHD) remains a major life threatening complication and one of the primary barriers to successful allogeneic hematopoietic stem cell transplantation, limiting its application in nonmalignant conditions. Immunosuppression is used for prevention and treatment of GVHD, dampening the graft versus leukemia effect. Intestinal bacteria play a major role in inflammation and augmenting the GVHD cytokine response. Early studies in murine models showed that manipulating the presence of intestinal flora or counteracting its byproducts could limit GVHD. Thus multiple clinical trials targeting gut decontamination were conducted, with the aims of modulating inflammation and protecting against GVHD, with mixed results. More recent work has improved our understanding of the role of intestinal microbiota in the maintenance of innate immunity, mucosal integrity and limiting inflammation. This review offers a summary of this data, with a discussion of potential therapeutic interventions manipulating the intestinal microbiota.

Keywords: Allogeneic hematopoietic stem cell transplantation, graft versus host disease, gut microbiota, *Lactobacillus*, Rifaximin.

1. INTRODUCTION

Allogeneic hematopoietic stem cell transplantation (allo-HSCT) is an important therapy for many malignant and nonmalignant disorders. Conditioning is the high-dose chemotherapy with or without total-body irradiation that precedes the infusion of allogeneic hematopoietic stem cells. The main purposes of conditioning are to kill malignant cells and to prevent graft rejection.

Acute graft versus host disease (GVHD) is the major toxicity of allo-HSCT and is a leading cause of transplant related mortality, limiting its wider application [1]. It is driven by donor T cells as they react against disparate host antigens. GVHD is responsible for 15-40% of transplant related mortality and is a major cause of morbidity [2]. It usually develops after myeloid engraftment, peaking in incidence three to four weeks post-transplant. It primarily affects three target organs: the gastrointestinal tract, the liver and the skin, causing varying degrees of diarrhea, jaundice, and rash. Prevention is the mainstay of GVHD management. Most patients receive calcineurin inhibitor based prophylaxis post-transplant [3]. Attempts to augment prophylaxis by utilizing ex-vivo T cell depletion or additional drugs may reduce the risk of developing GVHD, but, because of offsetting side effects (increased incidence of infection or relapse) have not led to improved survival [3]. Despite prophylaxis, moderate to severe GVHD develops in 30% or more of HLA matched related transplants and an even

greater percentage of T cell replete-unrelated donor transplants. Glucocorticoids are typically used to treat these patients. Although effective, this approach succeeds in only 50 to 60% of cases, and in severe cases usually fails, making severe GVHD fatal more than half the time [4]. Attempts at augmenting initial GVHD therapy by combining glucocorticoids with other agents, have, thus far, proven ineffective [5, 6].

Acute GVHD is often viewed as a tri-phasic process. The initial event is injury to the gastrointestinal mucosa (and to a lesser extent the skin) from the high dose chemotherapy and total body irradiation that are administered pre-transplant (phase I, recipient conditioning). The mucosal disruption permits bacterial translocation. In the lamina propria and mesenteric lymph nodes, translocated bacteria and bacterial products, particularly endotoxin, stimulate macrophages, which, in turn, secrete the pro-inflammatory cytokines TNF- α , IL-1 and IL-6. These cytokines enhance the activation of donor T cells by host antigen presenting cells (phase II, donor T cell activation). Gut injury also primes the inflammatory effector cells that are subsequently recruited by donor T cells (III, inflammatory effectors) [7]. Thus, the gut flora plays an integral role in the development of GVHD [8]. In this review, we will focus on the potential role of gut flora in the development of GVHD. We will summarize, in chronological order, relevant data supporting the role of bacteria in developing GVHD, and the trials that have attempted to abrogate the effect of gut flora on the development of GVHD through decontamination. We will then discuss properties of the gut microbiome that may offer insight into the pitfalls of the previous approach, and potential interventions.

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2. INTESTINAL MICROBIOTA AND GVHD: SEMINAL RESEARCH

The importance of the gut flora in the pathogenesis of GVHD was first recognized over forty years ago. In the early 1970s it was discovered that mice receiving allogeneic bone marrow after high dose total body irradiation could be protected from GVHD by raising them in germ free conditions or by "decontaminating" the gut with non-absorbable anti-bacterial antibiotics [9, 10].

2.1. Effect of Gut Injury Induced by Conditioning

Gut toxicity caused by conditioning results in disruption of the gut-mucosa barrier [11, 12], resulting in translocation of bacteria and endotoxin into the systemic circulation [13]. In murine BMT models, an increase in GVHD severity was seen after intensification of the conditioning regimen by increasing the dose of total body irradiation. This was mediated by systemic increases in TNF- α resulting from increased translocation of Lipopolysaccharide (LPS) into the systemic circulation [14]. LPS is a glucosamine-based phospholipid that makes up the outer monolayer of the outer membranes of most gram-negative bacteria. In murine GVHD models, administration of normally sublethal amounts of LPS during the early phase of acute GVHD triggers the release of significant amounts of TNF- α into the serum resulting in death. This is not evident in normal mice, consistent with experiments showing that macrophage priming by interferon is required for TNF production [15]. Endogenous LPS was detected in the serum of acute GVHD mice coincident with the onset of mortality. Countering the effect of the gut flora, by using an endotoxin antagonist, or the use of LPS resistant donor strains, diminished the incidence of GVHD [16, 17]. LPS alone may not trigger GVHD, but it amplifies the cytokine response involved [18].

In a study of 18 patients undergoing BMT, gastrointestinal permeability assessed by ^{51}Cr -EDTA absorption test increased significantly 2 days after the start of cytotoxic treatment, and remained elevated on days 1, 4, 7 and 10 post stem cell infusion, and correlated with cumulative gastrointestinal toxicity [12]. Additionally, the degree of clinical toxicity, and gut permeability correlated with acute GVHD severity [19, 20]. LPS is detected in the plasma of pediatric patients as early as day 0 of transplant [21].

2.2. The Effect of Gut Sterilization

Gut decontamination and other methods designed to counter the pro-inflammatory effects of the intestinal flora represent an alternative strategy to GVHD prophylaxis. Because its effects are predominantly local and anti-inflammatory, rather than systemic and immunosuppressive, it has the potential to more selectively prevent GVHD without interfering with engraftment, host immunity or the graft versus leukemia effect. This supposition is supported by murine studies, which demonstrated that countering the effect of the gut flora, using an endotoxin antagonist, diminished the incidence of GVHD, but did not affect graft versus leukemia activity [16]. The protective effect of gut decontamination on the risk of GVHD was reproduced in mice and dog transplant experiments [10, 22].

2.3. Clinical Studies in Gut Decontamination

The clinical impact of gastrointestinal decontamination was explored in the 1970s and 80s. Non-absorbable antibiotics (various combinations of vancomycin, gentamicin, polymyxin B and other antibiotics were used) were studied in combination with three other anti-microbial measures: a low microbial diet, aggressive skin cleansing and isolation in a laminar air flow room. Collectively, these measures were shown to reduce the incidence of GVHD, particularly in patients with aplastic anemia [23, 24].

There have been multiple studies since then addressing the impact of gut decontamination. A retrospective study of 198 adult matched sibling donor (MSD) transplant recipients who had been given oral nonabsorbable aminoglycosides demonstrated, using conventional culture methods that did not allow for specific identification of bacteria, that 24% achieved sustained decontamination for aerobic flora, 21% for anaerobic flora, and 15% had sustained decontamination for the entire bacterial flora. 23% of patients developed grade moderate to severe GVHD. Sustained growth suppression of anaerobic bacteria was associated with a lower incidence of acute GVHD (8% vs 30%, $p < 0.006$), while there was no significant association with growth suppression of aerobic bacteria [25].

A subsequent randomized controlled trial by the same group further highlighted the role of strictly anaerobic bacteria. In this study, oral metronidazole was administered from day -14 to day +35 in the randomized treatment arm, and all patients received oral ciprofloxacin. In the patients assigned to receive metronidazole in addition to ciprofloxacin the incidence of moderate to severe GVHD was 25%, while it was 50% in the patients receiving ciprofloxacin alone. Interestingly, metronidazole reduced the risk for skin and liver disease as well as gastrointestinal disease. This effect was observed only in patients receiving MSD transplantation [26].

Successful suppression of facultative and strictly anaerobic bacteria and yeast in the gut had a significant protective effect on the occurrence of GVHD in a study of 112 pediatric patients receiving myelablative MSD transplantation for hematologic malignancies. Gut decontamination was successful in 57/112 (51%). Nine patients developed GVHD (8 had mild and one patient had moderate disease), and patients with successful gut decontamination had significantly less GVHD ($p=0.013$). The one patient with moderate GVHD did not have successful gut decontamination. The fact that there were no cases of severe GVHD in this study may suggest that some degree of gut decontamination (though not complete) has a protective effect [27]. Yet another report suggested a role for gram-positive bacteria in promoting inflammation [28]. In addition, the results of a randomized controlled trial of fluconazole prophylaxis highlight the potential benefit of mitigating colonization of the gut by candida. In this study, patients assigned to fluconazole instead of placebo had lower incidences of death from severe gut GVHD as well as from invasive fungal infections [29]. Although the individual

contribution of administering non-absorbable antibiotics could not be gleaned from these studies, many transplant centers adopted and continue this practice today.

3. THE DYNAMIC RELATIONSHIP BETWEEN THE MICROBIOTA AND GUT IMMUNITY IN THE PATHOGENESIS OF GVHD

3.1. The role of Resident Bacterial Flora

The intestinal microbiota plays an important role in the maintenance of innate immunity as well as the protection against epithelial injury. One such example is through the effect of endogenous indole compounds on gut immune homeostasis. Commensal *Lactobacilli* utilize dietary tryptophan as an energy source and produce indole derivatives, such as indole-3-aldehyde that diffuses across the epithelial barrier and activates innate lymphoid cells (ILC) residing in the lamina propria of the gut. ILCs are critical to the development and regeneration of gut epithelium through local production of IL-22 that stimulates proliferation of epithelial stem cells and decreases ingress of pathogenic microbes [30, 31]. In humans, studies of commensal *Lactobacillus*, using different species combinations, show that *Lactobacillus* may stimulate the polarization of immune T cells toward regulatory T cells and have an anti-inflammatory effect in mouse colitis models [32, 33]. *Lactobacillus* can also decrease bacterial translocation from the intestine to mesenteric lymph nodes, and has a protective effect in maintaining the epithelial barrier by maintaining tight junctions [34, 35].

Intestinal flora can also stimulate secretion of antimicrobial proteins, such as α -defensins and Reg3 α , through the interaction of bacterial components and Toll-like and Nod receptors expressed on intestinal epithelial and Paneth cells [36, 37]. The role of these pathways is well established in inflammatory bowel disease [38]. In experimental GVHD models, administration of IL-22 could reduce intestinal pathology by enhancing secretion of peptide defensins from gut epithelial cells and by promoting the survival of intestinal stem cells [39]. And administration of ICA reduces GVHD and decreases bacterial colonization of mesenteric lymph nodes [40]. In patients undergoing allo-HSCT, plasma levels of Reg3 α , which primarily regulates gram-positive bacteria, correlate with severity of GI GVHD, and Paneth cell count in gut biopsies correlates with mortality [41, 42].

3.2. Loss of Bacterial Diversity

While studies in mice have shown reduction of GVHD with gut decontamination, clinical trials have shown mixed results. This raises the question whether an approach that indiscriminately eliminates all gut bacteria, including those that dampen the inflammatory response and promote epithelial integrity, is appropriate. More measured strategies may be more effective.

Determining the impact of antibiotics on the microbiota has long been hampered by the inability to culture most gut bacteria. The advent of deep sequencing of bacterial 16S

ribosomal RNA genes, however, has circumvented this problem [43]. Research using sequencing has made evident the potentially deleterious effects of antibiotics on the flora. This technology has revealed that in mice and humans ciprofloxacin, ampicillin, and other antibiotics not only reduce the overall population of fecal bacteria, they dramatically diminish diversity and this effect persists long after antibiotics have been discontinued [44, 45].

Murine studies have highlighted the potential of such perturbations of the microbiota to cause GVHD. Pre-transplant ampicillin, for instance, produced GVHD in one study. Its effects appeared to have been mediated by the loss of *Lactobacillales* from the flora. Interestingly, this pattern of change in flora was similar to that observed in stool samples obtained from patients with GVHD, and were not present in samples obtained from patients who did not develop this complication [46]. Conversely, the promotion of a diverse flora through use of probiotics can be protective. In the ampicillin study, the administration of *Lactobacillus johnsonii* (a species found as a commensal in humans and rodents), following antibiotics reversed ampicillin's effects. In another set of experiments, administration of *Lactobacillus rhamnosus* prevented GVHD by reducing translocation of enteric bacteria into the mesenteric lymph nodes [47].

Sequencing is also now being used clinically to assess the association between perturbations in the microbiota and GVHD. In a small study of 10 pediatric allo-HSCT patients, stool samples showed significant expansion of *Enterobacteriaceae* and decrease in small-chain fatty acid producing *Clostridia* in patient with intestinal GVHD compared to patients with GVHD, and that change preceded the clinical diagnosis [48]. In that same study, treatment with clindamycin, known to affect *Clostridia*, was significantly associated with development of GVHD. In another study, stool samples obtained at the time of engraftment from 90 adults undergoing allo-HSCT were analyzed, and patients divided into 3 groups according to bacterial diversity. Loss of diversity was linked to lower survival, increased transplant-related mortality, and more specifically death due to GVHD. The incidence of GVHD was not compared among the groups [49].

The loss of microbial diversity also puts allo-HSCT recipients at risk for infection, such as vancomycin resistant *Enterococcus* (VRE). Results of murine and clinical studies suggest that commensal bacteria belonging to the *Barnesiella* genus protects against VRE domination of the microbiota [46, 50]. VRE domination, in turn, has been shown to precede VRE bacteremia [45, 51].

4. STRATEGIES FOR THERAPEUTIC INTERVENTION

The imbalance in intestinal flora as a result of antibiotic exposure is to be expected in a patient population that has undergone intensive therapy prior to allo-HSCT. Restoration of diversity and reversing the antibiotic induced changes to the microbiota, may be an additional approach to protecting against GVHD.

4.1. Probiotics

The loss of *Lactobacillus* from the gut flora in allo-HSCT patients may play a role in promoting GVHD through adversely affecting immune tolerance, allowing *Enterococcus* expansion with subsequent worsening of intestinal inflammation, and possibly through loss of protective effect on other favorable gut microbiota.

Probiotics are broadly defined as live microorganisms which, when administered in adequate amounts, confer a health benefit on the host. Most common preparations include *Lactobacillus* either alone or in combination with other genera such as *Bifidobacterium*. Probiotics can inhibit the growth of pathogenic bacteria through mechanisms involving secretion of antimicrobial substances, strengthening the epithelial barrier and modulation of the immune system [52]. Thus they can help maintain bacterial diversity within the gut. In rats treated with azithromycin, the group treated with *Lactobacillus* had a higher number of *Bacteroides* and lower number of *Enterobacteriaceae* compared to controls [53]. In a placebo controlled study of healthy adults treated with clindamycin, supplementation with *Lactobacillus* prevented disturbances in the numbers of intestinal *Bacteroides* [54].

The use of probiotics is promoted in the prevention of antibiotic induced diarrhea, and in the treatment of inflammatory bowel disease (IBD), irritable bowel syndrome (IBS), and necrotizing enterocolitis in premature infants, and supported by the results of multiple randomized controlled trials [55-57]. One study in adults with cancer receiving pelvic radiation reported a decrease in diarrhea compared to placebo [58]. Importantly, no adverse events were reported in this immunocompromised population, consistent with the safety profile in other published trials [59]. However, rare reports of bacteremia attributed to probiotic use have emerged, one in a patient who underwent allo-HSCT [60]. A few ongoing clinical trials, in pediatrics and adults, are examining the safety of probiotic (*Lactobacillus*) administration in stem cell transplantation, and beginning to investigate its effects on infection as well as GVHD.

4.2. Rifaximin

Rifaximin, is a non-absorbable rifamycin derivative that is used as an antibiotic in the treatment of several gastrointestinal diseases, such as hepatic encephalopathy, small intestine bacterial overgrowth, IBD and colonic diverticular disease [61-64]. It has a low toxicity profile, and is well tolerated in children and adults [65].

It has biologic properties that could make it more effective than other antibiotics for countering the pro-GVHD effects of intestinal microbiota. It is a bacteriostatic agent that binds the beta-subunit of bacterial DNA-dependent RNA polymerase and, thereby, inhibits bacterial RNA synthesis [65]. That it is bacteriostatic could be advantageous, since bactericidal agents, but not bacteriostatic agents, induce the release of free endotoxin, and thereby promote inflammation [66-68]. It has very broad activity, covering both gram-positive and gram-negative bacteria and aerobes as well as anaerobes [69, 70]. Rifaximin inhibits bacterial

translocation. This was demonstrated in a study using a murine model of colitis. In this study rifaximin reduced the recovery of bacteria from mesenteric lymph nodes, diminished the secretion of IL-2, IFN- γ and TNF- α and hastened healing [71]. Rifaximin also exhibits anti-inflammatory effects in pre-clinical models of IBD [72, 73]. In a model of chronic water avoidance stress (a visceral hyperalgesia model that is similar to human IBS), rats developed mucosal inflammation with increase in IL-17, IL-6, and TNF α gene expression in the distal ileal tissue, an increase in gut permeability, and decrease bacterial diversity. Treatment with rifaximin decreased the IL-17, IL-1 β , Interferon- γ and TNF- α mRNA levels and increased tight junction occluding mRNA level [74]. Finally, rifaximin does not appear to adversely alter intestinal microbiota diversity [75]. In fact, in one murine study, rifaximin treatment increased the proportion of *Lactobacillus* species [76].

The proposed mechanisms behind rifaximin's efficacy are relevant to GVHD. In a safety and feasibility trial, we administered rifaximin to 20 adolescent and adult allo-HSCT patients receiving intensive conditioning. Compared to historical controls, the early increase in the treatment group's mean IL-6 level was 73% less than that in the control group, suggesting an effect on conditioning-induced inflammation. We did not analyze effects on intestinal flora. In that limited trial, there was no difference in the occurrence of GVHD [77]. Further studies are needed to elucidate the effects of rifaximin. Most patients undergoing allo-HSCT receive broad spectrum antibiotics for prophylaxis and/or treatment of infection, and while institutional approaches may change towards prophylaxis, coverage with broad spectrum antibiotics after the development of fever is standard practice, given the high risk of bacteremia and mortality in this patient population [78-80]. Combining standard antibiotics with a probiotic or with rifaximin may serve the dual role of protection against bacteremia and restoring bacterial balance within the gut.

SUMMARY

The intestinal microbiota is crucial in the pathogenesis of GVHD. Recent deep sequencing methods have enabled researchers to delve deep into the complex relationship between the microbiota and GVHD. Favorably modulating the intestinal flora may abrogate their pro-inflammatory effects and could aid in the prevention of GVHD, as an adjunct to anti-T cell agents. Through data gleaned from other inflammatory conditions affecting the gut, rifaximin may emerge as a potential agent that could restore favorable bacterial balance, and impact risk of GVHD, without augmenting immune suppression, thus leading to overall improved outcomes in allo-HSCT.

CONFLICT OF INTEREST

The author(s) confirm that this article content has no conflict of interest.

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REFERENCES

- [1] Ferrara, J. L.; Reddy, P.; Pathophysiology of graft-versus-host disease. *Semin Hematol.*, **2006**, 43(1), 3-10.
- [2] Appelbaum, F. R.; Haematopoietic cell transplantation as immunotherapy. *Nature*, **2001**, 411(6835), 385-389.
- [3] Sullivan, K.; Graft -vs.-Host Disease. In: *Thomas' Hematopoietic Cell Transplantation*; Appelbaum, Ed.; Blackwell Publishing Ltd.: Oxford, UK, **2004**. p 635-664.
- [4] Przepiorka, D.; Weisdorf, D.; Martin, P.; Klingemann, H. G.; Beatty, P.; Hows, J.; Thomas, E. D.; 1994 Consensus Conference on Acute GVHD Grading. *Bone Marrow Transplant*, **1995**, 15(6), 825-828.
- [5] Cragg, L.; Blazar, B. R.; Defor, T.; Kolatker, N.; Miller, W.; Kersey, J.; Ramsay, M.; McGlave, P.; Filipovich, A.; Weisdorf, D.; A randomized trial comparing prednisone with antithymocyte globulin/prednisone as an initial systemic therapy for moderately severe acute graft-versus-host disease. *Biol. Blood Marrow Transplant*, **2000**, 6(4A), 441-447.
- [6] Lee, S. J.; Zahrieh, D.; Agura, E.; MacMillan, M. L.; Maziarz, R. T.; McCarthy, P. L., Jr.; Ho, V. T.; Cutler, C.; Alyea, E. P.; Antin, J. H.; Soiffer, R. J.; Effect of up-front daclizumab when combined with steroids for the treatment of acute graft-versus-host disease: results of a randomized trial. *Blood*, **2004**, 104(5), 1559-1564.
- [7] Ferrara, J. L. M.; Antin, J.; The Pathophysiology of Graft -vs.-Host Disease. In: *Thomas' Hematopoietic Cell Transplantation*; Appelbaum, Ed.; Blackwell Publishing, 3rd ed. Oxford, **2004**.
- [8] MacMillan, M. L.; DeFor, T. E.; Weisdorf, D. J.; What predicts high risk acute graft-versus-host disease (GVHD) at onset?: identification of those at highest risk by a novel acute GVHD risk score. *Br. J. Haematol.*, **2012**, 157(6), 732-741.
- [9] Jones, J. M.; Wilson, R.; Bealmeier, P. M.; Mortality and gross pathology of secondary disease in germfree mouse radiation chimeras. *Radiat. Res.*, **1971**, 45(3), 577-588.
- [10] van Bekkum, D. W.; Roodenburg, J.; Heidt, P. J.; van der Waaij, D.; Mitigation of secondary disease of allogeneic mouse radiation chimeras by modification of the intestinal microflora. *J. Natl. Cancer Inst.*, **1974**, 52(2), 401-404.
- [11] Fegan, C.; Poynton, C. H.; Whittaker, J. A.; The gut mucosal barrier in bone marrow transplantation. *Bone Marrow Transplant*, **1990**, 5(6), 373-377.
- [12] Johansson, J. E.; Ekman, T.; Gastro-intestinal toxicity related to bone marrow transplantation: disruption of the intestinal barrier precedes clinical findings. *Bone Marrow Transplant*, **1997**, 19(9), 921-925.
- [13] Berg, R.; Bacterial Translocation from the Gastrointestinal Tracts of Mice Receiving Immunosuppressive Chemotherapeutic Agents. *Curr. Microbiol.*, **1983**, 8(5), 285-292.
- [14] Hill, G. R.; Crawford, J. M.; Cooke, K. R.; Brinson, Y. S.; Pan, L.; Ferrara, J. L.; Total body irradiation and acute graft-versus-host disease: the role of gastrointestinal damage and inflammatory cytokines. *Blood*, **1997**, 90(8), 3204-3213.
- [15] Nestel, F. P.; Price, K. S.; Seemayer, T. A.; Lapp, W. S.; Macrophage priming and lipopolysaccharide-triggered release of tumor necrosis factor alpha during graft-versus-host disease. *J. Exp. Med.*, **1992**, 175(2), 405-413.
- [16] Cooke, K. R.; Gerbitz, A.; Crawford, J. M.; Teshima, T.; Hill, G. R.; Tesolin, A.; Rossignol, D. P.; Ferrara, J. L.; LPS antagonism reduces graft-versus-host disease and preserves graft-versus-leukemia activity after experimental bone marrow transplantation. *J. Clin. Invest.*, **2001**, 107(12), 1581-1589.
- [17] Cooke, K. R.; Hill, G. R.; Crawford, J. M.; Bungard, D.; Brinson, Y. S.; Delmonte, J., Jr.; Ferrara, J. L.; Tumor necrosis factor- α production to lipopolysaccharide stimulation by donor cells predicts the severity of experimental acute graft-versus-host disease. *J. Clin. Invest.*, **1998**, 102(10), 1882-1891.
- [18] Lampert, I. A.; Moore, R. H.; Huby, R.; Cohen, J.; Observations on the role of endotoxin in graft-versus-host disease. *Prog. Clin. Biol. Res.*, **1988**, 272, 351-359.
- [19] Goldberg, J.; Jacobsohn, D. A.; Zahurak, M. L.; Vogelsang, G. B.; Gastrointestinal toxicity from the preparative regimen is associated with an increased risk of graft-versus-host disease. *Biol. Blood Marrow Transplant*, **2005**, 11(2), 101-107.
- [20] Johansson, J. E.; Ekman, T.; Gut toxicity during hemopoietic stem cell transplantation may predict acute graft-versus-host disease severity in patients. *Digest. Dis. Sci.*, **2007**, 52(9), 2340-2345.
- [21] Levy, O.; Teixeira-Pinto, A.; White, M. L.; Carroll, S. F.; Lehmann, L.; Wypij, D.; Guinan, E.; Endotoxemia and elevation of lipopolysaccharide-binding protein after hematopoietic stem cell transplantation. *Pediat. Infect. Dis. J.*, **2003**, 22(11), 978-981.
- [22] Vriesendorp, H. M.; Heidt, P. J.; Zurcher, C.; Gastrointestinal decontamination of dogs treated with total body irradiation and bone marrow transplantation. *Exp. Hematol.*, **1981**, 9(9), 904-916.
- [23] Buckner, C. D.; Clift, R. A.; Sanders, J. E.; Meyers, J. D.; Counts, G. W.; Farewell, V. T.; Thomas, E. D.; Protective environment for marrow transplant recipients: a prospective study. *Ann. Intern. Med.*, **1978**, 89(6), 893-901.
- [24] Storb, R.; Prentice, R. L.; Buckner, C. D.; Clift, R. A.; Appelbaum, F.; Deeg, J.; Doney, K.; Hansen, J. A.; Mason, M.; Sanders, J. E.; Singer, J.; Sullivan, K. M.; Witherspoon, R. P.; Thomas, E. D.; Graft-versus-host disease and survival in patients with aplastic anemia treated by marrow grafts from HLA-identical siblings. Beneficial effect of a protective environment. *N. Engl. J. Med.*, **1983**, 308(6), 302-307.
- [25] Beelen, D. W.; Haralambie, E.; Brandt, H.; Linzenmeier, G.; Muller, K. D.; Quabeck, K.; Sayer, H. G.; Graeven, U.; Mahmoud, H. K.; Schaefer, U. W.; Evidence that sustained growth suppression of intestinal anaerobic bacteria reduces the risk of acute graft-versus-host disease after sibling marrow transplantation. *Blood*, **1992**, 80(10), 2668-2676.
- [26] Beelen, D. W.; Elmaagacli, A.; Muller, K. D.; Hirche, H.; Schaefer, U. W.; Influence of intestinal bacterial decontamination using metronidazole and ciprofloxacin or ciprofloxacin alone on the development of acute graft-versus-host disease after marrow transplantation in patients with hematologic malignancies: final results and long-term follow-up of an open-label prospective randomized trial. *Blood*, **1999**, 93(10), 3267-3275.
- [27] Vossen, J. M.; Guiot, H. F.; Lankester, A. C.; Vossen, A. C.; Bredius, R. G.; Wolterbeek, R.; Bakker, H. D.; Heidt, P. J.; Complete suppression of the gut microbiome prevents acute graft-versus-host disease following allogeneic bone marrow transplantation. *PLoS one*, **2014**, 9(9), e105706.
- [28] Holler, E.; Rogler, G.; Brenmoehl, J.; Hahn, J.; Herfarth, H.; Greinix, H.; Dickinson, A. M.; Socie, G.; Wolff, D.; Fischer, G.; Jackson, G.; Rocha, V.; Steiner, B.; Eissner, G.; Marienhagen, J.; Scholmerich, J.; Andreesen, R.; Prognostic significance of NOD2/CARD15 variants in HLA-identical sibling hematopoietic stem cell transplantation: effect on long-term outcome is confirmed in 2 independent cohorts and may be modulated by the type of gastrointestinal decontamination. *Blood*, **2006**, 107(10), 4189-4193.
- [29] Marr, K. A.; Seidel, K.; Slavin, M. A.; Bowden, R. A.; Schoch, H. G.; Flowers, M. E.; Corey, L.; Boeckh, M.; Prolonged fluconazole prophylaxis is associated with persistent protection against candidiasis-related death in allogeneic marrow transplant recipients: long-term follow-up of a randomized, placebo-controlled trial. *Blood*, **2000**, 96(6), 2055-2061.
- [30] Behnsen, J.; Raffatellu, M.; Keeping the peace: aryl hydrocarbon receptor signaling modulates the mucosal microbiota. *Immunity*, **2013**, 39(2), 206-207.
- [31] Zelante, T.; Iannitti, R. G.; Cunha, C.; De Luca, A.; Giovannini, G.; Pieraccini, G.; Zecchi, R.; D'Angelo, C.; Massi-Benedetti, C.; Fallarino, F.; Carvalho, A.; Puccetti, P.; Romani, L.; Tryptophan catabolites from microbiota engage aryl hydrocarbon receptor and balance mucosal reactivity via interleukin-22. *Immunity*, **2013**, 39(2), 372-385.
- [32] van Baaren, P.; Troost, F. J.; van Hemert, S.; van der Meer, C.; de Vos, W. M.; de Groot, P. J.; Hooiveld, G. J.; Brummer, R. J.; Kleerebezem, M.; Differential NF-kappaB pathways induction by *Lactobacillus plantarum* in the duodenum of healthy humans correlating with immune tolerance. *Proc. Nat. Acad. Sci. U.S.A.*, **2009**, 106(7), 2371-2376.
- [33] Toumi, R.; Soufli, I.; Rafa, H.; Belkhef, M.; Biad, A.; Touil-Boukoffa, C.; Probiotic bacteria *Lactobacillus* and *Bifidobacterium* attenuate inflammation in dextran sulfate sodium-induced experimental colitis in mice. *Intl. J. Immunopathol. Pharmacol.*, **2014**, 27(4), 615-627.
- [34] Dicksved, J.; Schreiber, O.; Willing, B.; Petersson, J.; Rang, S.; Phillipson, M.; Holm, L.; Roos, S.; *Lactobacillus reuteri* maintains

- a functional mucosal barrier during DSS treatment despite mucus layer dysfunction. *PLoS One*, **2012**, 7(9), e46399.
- [35] Mennigen, R.; Nolte, K.; Rijcken, E.; Utech, M.; Loeffler, B.; Senninger, N.; Bruewer, M.; Probiotic mixture VSL#3 protects the epithelial barrier by maintaining tight junction protein expression and preventing apoptosis in a murine model of colitis. *Amer. J. physiol. Gastrointest. Liver Physiol.*, **2009**, 296(5), G1140-1149.
- [36] Ayabe, T.; Satchell, D. P.; Wilson, C. L.; Parks, W. C.; Selsted, M. E.; Ouellette, A. J.; Secretion of microbicidal alpha-defensins by intestinal Paneth cells in response to bacteria. *Nat. Immunol.*, **2000**, 1(2), 113-118.
- [37] Pamer, E. G.; Immune responses to commensal and environmental microbes. *Nat. Immunol.*, **2007**, 8(11), 1173-1178.
- [38] Adachi, J. A.; DuPont, H. L.; Rifaximin: a novel nonabsorbed rifamycin for gastrointestinal disorders. *Clin. Infect. Dis.*, **2006**, 42(4), 541-547.
- [39] Lindemann, C.; Mertelsmann, A.; Dudakov, J. A.; Velardi, E.; Hua, G.; O'Connor, M.; Kolesnick, R.; van den Brink, M. R. M.; Hanash, A. M.; IL-22 Administration Protects Intestinal Stem Cells from Gvhd. *Biol. Blood Marrow Transplant.*, **2014**, 20(2, Supplement), S53-S54.
- [40] Waller, E. K.; Giver, C. R.; Rangaraju, S.; Swimm, A.; Sharma, A.; Qayed, M.; Defilipp, Z.; Kalman, D. Administration of a Tryptophan Metabolite, Indole-3-Carboxaldehyde, Reduces Graft Versus Host Disease Morbidity and Mortality and Enhances Gastrointestinal Barrier Function in a Murine Model of Allogeneic Bone Marrow Transplantation; **2014**.
- [41] Eriguchi, Y.; Takashima, S.; Oka, H.; Shimoji, S.; Nakamura, K.; Uryu, H.; Shimoda, S.; Iwasaki, H.; Shimonon, N.; Ayabe, T.; Akashi, K.; Teshima, T.; Graft-versus-host disease disrupts intestinal microbial ecology by inhibiting Paneth cell production of alpha-defensins. *Blood*, **2012**, 120(1), 223-231.
- [42] Ferrara, J. L.; Harris, A. C.; Greenson, J. K.; Braun, T. M.; Holler, E.; Teshima, T.; Levine, J. E.; Choi, S. W.; Huber, E.; Landfried, K.; Akashi, K.; Vander Lugt, M.; Reddy, P.; Chin, A.; Zhang, Q.; Hanash, S.; Paczesny, S.; Regenerating islet-derived 3-alpha is a biomarker of gastrointestinal graft-versus-host disease. *Blood*, **2011**, 118(25), 6702-6708.
- [43] Eckburg, P. B.; Bik, E. M.; Bernstein, C. N.; Purdom, E.; Dethlefsen, L.; Sargent, M.; Gill, S. R.; Nelson, K. E.; Relman, D. A.; Diversity of the human intestinal microbial flora. *Science*, **2005**, 308(5728), 1635-1638.
- [44] Dethlefsen, L.; Huse, S.; Sogin, M. L.; Relman, D. A.; The pervasive effects of an antibiotic on the human gut microbiota, as revealed by deep 16S rRNA sequencing. *PLoS Biology*, **2008**, 6(11), e280.
- [45] Ubeda, C.; Taur, Y.; Jenq, R. R.; Equinda, M. J.; Son, T.; Samstein, M.; Viale, A.; Soccì, N. D.; van den Brink, M. R.; Kamboj, M.; Pamer, E. G.; Vancomycin-resistant *Enterococcus* domination of intestinal microbiota is enabled by antibiotic treatment in mice and precedes bloodstream invasion in humans. *J. Clin. Invest.*, **2010**, 120(12), 4332-4341.
- [46] Jenq, R. R.; Ubeda, C.; Taur, Y.; Menezes, C. C.; Khanin, R.; Dudakov, J. A.; Liu, C.; West, M. L.; Singer, N. V.; Equinda, M. J.; Gobourne, A.; Lipuma, L.; Young, L. F.; Smith, O. M.; Ghosh, A.; Hanash, A. M.; Goldberg, J. D.; Aoyama, K.; Blazar, B. R.; Pamer, E. G.; van den Brink, M. R.; Regulation of intestinal inflammation by microbiota following allogeneic bone marrow transplantation. *J. Exp. Med.*, **2012**, 209(5), 903-911.
- [47] Gerbitz, A.; Schultz, M.; Wilke, A.; Linde, H. J.; Scholmerich, J.; Andreesen, R.; Holler, E.; Probiotic effects on experimental graft-versus-host disease: let them eat yogurt. *Blood*, **2004**, 103(11), 4365-4367.
- [48] Simms-Waldrip, T.; Meir, M.; Fan, D.; Coughlin, L.; Savani, M.; Watt, T.; Aquino, V.; Young Koh, V.; The Role of Gut Microbiota in the Development of Intestinal GVHD. *Biol. Blood Marrow Transplant.*, **2014**, 20(2), S55-S56.
- [49] Taur, Y.; Xavier, J. B.; Lipuma, L.; Ubeda, C.; Goldberg, J.; Gobourne, A.; Lee, Y. J.; Dubin, K. A.; Soccì, N. D.; Viale, A.; Perales, M. A.; Jenq, R. R.; van den Brink, M. R.; Pamer, E. G.; Intestinal domination and the risk of bacteremia in patients undergoing allogeneic hematopoietic stem cell transplantation. *Clin. Infect. Dis.*, **2012**, 55(7), 905-914.
- [50] Ubeda, C.; Bucci, V.; Caballero, S.; Djukovic, A.; Toussaint, N. C.; Equinda, M.; Lipuma, L.; Ling, L.; Gobourne, A.; No, D.; Taur, Y.; Jenq, R. R.; van den Brink, M. R.; Xavier, J. B.; Pamer, E. G.; Intestinal microbiota containing *Barnesiella* species cures vancomycin-resistant *Enterococcus faecium* colonization. *Infect. Immun.*, **2013**, 81(3), 965-973.
- [51] Taur, Y.; Jenq, R. R.; Perales, M. A.; Littmann, E. R.; Morjaria, S.; Ling, L.; No, D.; Gobourne, A.; Viale, A.; Dahi, P. B.; Ponce, D. M.; Barker, J. N.; Giral, S.; van den Brink, M.; Pamer, E. G.; The effects of intestinal tract bacterial diversity on mortality following allogeneic hematopoietic stem cell transplantation. *Blood*, **2014**, 124(7), 1174-1182.
- [52] Bermudez-Brito, M.; Plaza-Diaz, J.; Munoz-Quezada, S.; Gomez-Llorente, C.; Gil, A.; Probiotic mechanisms of action. *Ann. Nutr. Metabol.*, **2012**, 61(2), 160-174.
- [53] Shoaib, A.; Dachang, W.; Xin, Y.; Determining the role of a probiotic in the restoration of intestinal microbial balance by molecular and cultural techniques. *Gene. Mol. Res.: GMR*, **2015**, 14(1), 1526-1537.
- [54] Sullivan, A.; Barkholt, L.; Nord, C. E.; *Lactobacillus acidophilus*, *Bifidobacterium lactis* and *Lactobacillus F19* prevent antibiotic-associated ecological disturbances of *Bacteroides fragilis* in the intestine. *J. Antimicrob. Chemother.*, **2003**, 52(2), 308-311.
- [55] Alfaleh, K.; Anabrees, J.; Bassler, D.; Al-Kharfi, T.; Probiotics for prevention of necrotizing enterocolitis in preterm infants. *Cochrane. Database Syst. Rev.*, **2011**, 16(3), CD005496.
- [56] Ghouri, Y. A.; Richards, D. M.; Rahimi, E. F.; Krill, J. T.; Jelinek, K. A.; DuPont, A. W.; Systematic review of randomized controlled trials of probiotics, prebiotics, and synbiotics in inflammatory bowel disease. *Clin. Exp. Gastroenterol.*, **2014**, 7, 473-487.
- [57] Moayyedi, P.; Ford, A. C.; Talley, N. J.; Cremonini, F.; Fox-Orenstein, A. E.; Brandt, L. J.; Quigley, E. M.; The efficacy of probiotics in the treatment of irritable bowel syndrome: a systematic review. *Gut*, **2010**, 59(3), 325-332.
- [58] Delia, P.; Sansotta, G.; Donato, V.; Frosina, P.; Messina, G.; De Renzis, C.; Famularo, G.; Use of probiotics for prevention of radiation-induced diarrhea. *World J. Gastroenterol.: WJG*, **2007**, 13(6), 912-915.
- [59] Hempel, S.; Newberry, S.; Ruelaz, A.; Wang, Z.; Miles, J. N.; Suttrop, M. J.; Johnsen, B.; Shanman, R.; Slusser, W.; Fu, N.; Smith, A.; Roth, B.; Polak, J.; Motala, A.; Perry, T.; Shekelle, P. G.; Safety of probiotics used to reduce risk and prevent or treat disease. *Evid. Rep. Tech. Assess.*, **2011**, (200), 1-645.
- [60] Mehta, A.; Rangarajan, S.; Borate, U.; A cautionary tale for probiotic use in hematopoietic SCT patients-*Lactobacillus acidophilus* sepsis in a patient with mantle cell lymphoma undergoing hematopoietic SCT. *Bone Marrow Transplant*, **2013**, 48(3), 461-462.
- [61] DuPont, H. L.; Jiang, Z. D.; Okhuysen, P. C.; Ericsson, C. D.; de la Cabada, F. J.; Ke, S.; DuPont, M. W.; Martinez-Sandoval, F.; A randomized, double-blind, placebo-controlled trial of rifaximin to prevent travelers' diarrhea. *Ann. Intern. Med.*, **2005**, 142(10), 805-812.
- [62] Miglio, F.; Valpiani, D.; Rossellini, S. R.; Ferrieri, A.; Rifaximin, a non-absorbable rifamycin, for the treatment of hepatic encephalopathy. A double-blind, randomised trial. *Curr. Med. Res. Opin.*, **1997**, 13(10), 593-601.
- [63] Bucci, L.; Palmieri, G. C.; Double-blind, double-dummy comparison between treatment with rifaximin and lactulose in patients with medium to severe degree hepatic encephalopathy. *Curr. Med. Res. Opin.*, **1993**, 13(2), 109-118.
- [64] Lauritano, E. C.; Gabrielli, M.; Lupascu, A.; Santoliquido, A.; Nucera, G.; Scarpellini, E.; Vincenti, F.; Cammarota, G.; Flore, R.; Pola, P.; Gasbarrini, G.; Gasbarrini, A.; Rifaximin dose-finding study for the treatment of small intestinal bacterial overgrowth. *Aliment. Pharmacol. Ther.*, **2005**, 22(1), 31-35.
- [65] Ojetti, V.; Lauritano, E. C.; Barbaro, F.; Migneco, A.; Ainora, M. E.; Fontana, L.; Gabrielli, M.; Gasbarrini, A.; Rifaximin pharmacology and clinical implications. *Exp. Opin. Drug Metabol. Toxicol.*, **2009**, 5(6), 675-682.
- [66] Goto, H.; Nakamura, S.; Liberation of endotoxin from *Escherichia coli* by addition of antibiotics. *Jpn. J. Exp. Med.*, **1980**, 50(1), 35-43.
- [67] Shenep, J. L.; Barton, R. P.; Mogan, K. A.; Role of antibiotic class in the rate of liberation of endotoxin during therapy for experimental gram-negative bacterial sepsis. *J. Infect. Dis.*, **1985**, 151(6), 1012-1018.

- [68] Dofferhoff, A. S.; Esselink, M. T.; de Vries-Hospers, H. G.; van Zanten, A.; Bom, V. J.; Weits, J.; Vellenga, E.; The release of endotoxin from antibiotic-treated *Escherichia coli* and the production of tumour necrosis factor by human monocytes. *J. Antimicrob. Chemother.*, **1993**, 31(3), 373-384.
- [69] Hoover, W. W.; Gerlach, E. H.; Hoban, D. J.; Eliopoulos, G. M.; Pfaller, M. A.; Jones, R. N.; Antimicrobial activity and spectrum of rifaximin, a new topical rifamycin derivative. *Diagn. Microbiol. Infect. Dis.*, **1993**, 16(2), 111-118.
- [70] Gillis, J. C.; Brogden, R. N.; Rifaximin. A review of its antibacterial activity, pharmacokinetic properties and therapeutic potential in conditions mediated by gastrointestinal bacteria. *Drugs*, **1995**, 49(3), 467-484.
- [71] Fiorucci, S.; Distrutti, E.; Mencarelli, A.; Barbanti, M.; Palazzini, E.; Morelli, A.; Inhibition of intestinal bacterial translocation with rifaximin modulates lamina propria monocytic cells reactivity and protects against inflammation in a rodent model of colitis. *Digestion*, **2002**, 66(4), 246-256.
- [72] Cheng, J.; Shah, Y. M.; Ma, X.; Pang, X.; Tanaka, T.; Kodama, T.; Krausz, K. W.; Gonzalez, F. J.; Therapeutic role of rifaximin in inflammatory bowel disease: clinical implication of human pregnane X receptor activation. *J. Pharmacol. Experiment. Therapeut.*, **2010**, 335(1), 32-41.
- [73] Shono, Y.; Tuckett, A. Z.; Ouk, S.; Liou, H. C.; Altan-Bonnet, G.; Tsai, J. J.; Oyler, J. E.; Smith, O. M.; West, M. L.; Singer, N. V.; Doubrovina, E.; Pankov, D.; Undhad, C. V.; Murphy, G. F.; Lezcano, C.; Liu, C.; O'Reilly, R. J.; van den Brink, M. R.; Zakrzewski, J. L.; A small-molecule c-Rel inhibitor reduces alloactivation of T cells without compromising antitumor activity. *Cancer Discov.*, **2014**, 4(5), 578-591.
- [74] Xu, D.; Gao, J.; Gilliland, M., 3rd; Wu, X.; Song, I.; Kao, J. Y.; Owyang, C.; Rifaximin alters intestinal bacteria and prevents stress-induced gut inflammation and visceral hyperalgesia in rats. *Gastroenterology*, **2014**, 146(2), 484-496 e484.
- [75] Kakiyama, G.; Pandak, W. M.; Gillevet, P. M.; Hylemon, P. B.; Heuman, D. M.; Daita, K.; Takei, H.; Muto, A.; Nittono, H.; Ridlon, J. M.; White, M. B.; Noble, N. A.; Monteith, P.; Fuchs, M.; Thacker, L. R.; Sikaroodi, M.; Bajaj, J. S.; Modulation of the fecal bile acid profile by gut microbiota in cirrhosis. *J. Hepatol.*, **2013**, 58(5), 949-955.
- [76] Gao, J.; Gilliland, M. G., 3rd; Owyang, C.; Rifaximin, gut microbes and mucosal inflammation: unraveling a complex relationship. *Gut Microbes*, **2014**, 5(4), 571-575.
- [77] Qayed, M.; Langston, A.; Chiang, K. Y.; August, K.; Hilinski, J. A.; Cole, C. R.; Rogatko, A.; Bostick, R. M.; Horan, J. T.; Rifaximin for preventing acute graft-versus-host disease: impact on plasma markers of inflammation and T-cell activation. *J. Pediatr. Hematol. Oncol.*, **2013**, 35(4), e149-152.
- [78] Blennow, O.; Ljungman, P.; Sparrelid, E.; Mattsson, J.; Remberger, M.; Incidence, risk factors, and outcome of bloodstream infections during the pre-engraftment phase in 521 allogeneic hematopoietic stem cell transplantations. *Transplant Infectious Disease : J. Transplant. Soc.*, **2014**, 16(1), 106-114.
- [79] Mikulska, M.; Del Bono, V.; Bruzzi, P.; Raiola, A. M.; Gualandi, F.; Van Lint, M. T.; Bacigalupo, A.; Viscoli, C.; Mortality after bloodstream infections in allogeneic haematopoietic stem cell transplant (HSCT) recipients. *Infection*, **2012**, 40(3), 271-278.
- [80] Simonsen, K. A.; Reed, M. P.; Mabry, M. S.; Zhang, Y.; Longo, W. L.; Retrospective analysis of fluoroquinolone prophylaxis in patients undergoing allogeneic hematopoietic stem cell transplantation. *J. Oncol. Pharm. Pract.*, **2013**, 19(4), 291-297.